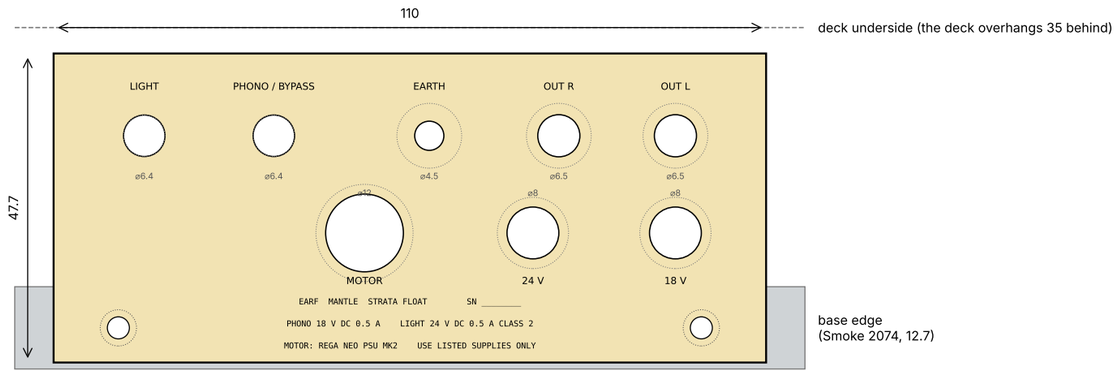
Mantle: Strata Float v3, rear plate, wiring and earth, rating, the light (round 48)

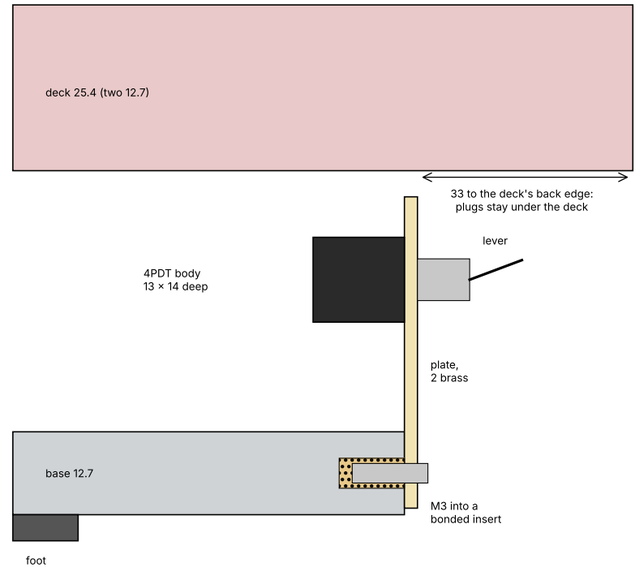
mm. A proposal until the owner signs it. The audit's Strata items (ENGINEERING-AUDIT.md, r46-owner-notes): no rear panel, no wiring or earth drawing, no rating label; an 8 mm strip in a 5 mm slot with 2 mm corners; a 25 W supply. The light moved into the deck on 2026-11-03.

A Rear plate, seen from behind (2:1 on the sheet), 2 mm lead-free brass, brushed; legends and rating engraved and filled black

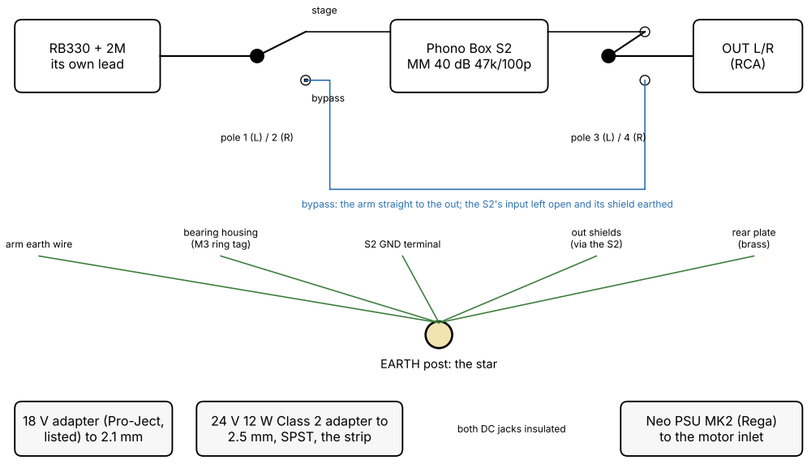


Two M3 × 10 button screws into brass inserts bonded in the base's edge (4.6, 10 deep). Holes from the CAD (mantle_strata_v3.REAR_HW); rating 2 mm caps, 3 lines, between the screws. No compliance mark until tested; the supplies carry their own.

B Section on the 4PDT (1:1): the plate under the overhang



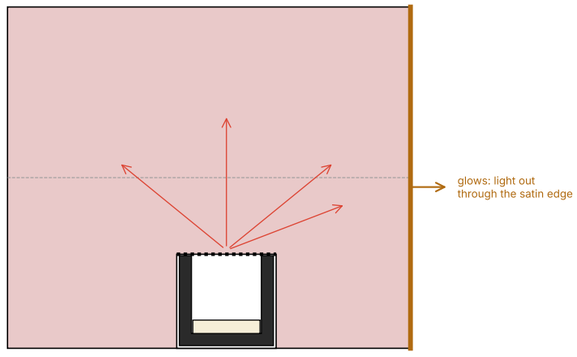
C Signal, earth and power (one channel drawn; the other the same)



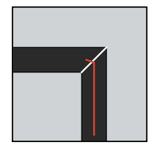
Rules: one earth point (the post), no loops; the light's 24 V and the motor's lead never share a ground with signal, and run along the base's far side from the arm's lead (40 or more apart, crossing at right angles); shielded leads, shields earthed at the S2 end only. The two DC jacks are insulated from the plate (the S2's 0 V reaches the star only through its GND terminal).

D The light, the deck edge-lit: section through the groove (4:1) and a corner, plan (1:1)

deck, 25.4 (two 12.7, bond line dashed); the groove 7.4 × 7 from below, 10 in from the edge, its ceiling bead-blasted (dotted) so the light spreads into the slab



U-channel 7 × 7, black anodized, open face up; the strip 5 mm on its floor, facing up. Below, only the channel's black floor shows: nothing falls on the shelf but a faint spill.



a corner from below
(in the deck):
the channel mitred at 45,
the strip's runs joined by
a soldered jumper (red):
COB strip does not bend
in its own plane

The light: 1.51 m of 5 mm 24 V COB strip at about 4 W/m (5 W/m at most) is about 6.0 W (7.6 W at most), 0.25 A, on a listed Class 2 / LPS 24 V adapter of 12 W: under half its rating, and a PS1 circuit under UL 62368-1. No PWM dimmer inside (its FCC Part 15B status was open): one brightness, set by the strip's choice, and an SPST on the plate. If the owner wants it dimmable, a listed Class 2 dimmer in the adapter's lead, outside the record player, keeps the plate free of clocked circuits. First-sample checks: hum and noise at the out in both switch positions, into an outside phono stage, with the light on and off and the motor running (A-weighted, against the S2's own figure); the 2M's frequency response in bypass, into 47k, against the stage position (no droop above 10 kHz); continuity and isolation of the two DC jacks from the plate.